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Mandelbrot Upic

A Commodore 64 Ultimate demo that generates a Mandelbrot fractal on-device at 64 MHz turbo, packs it directly into Upic format (a 16-color, 384x256 border-color raster picture technique), and displays it live.

Spun off from [landofficeandfire](#), which built and hardware-validated the Upic viewer this project reuses. See CREDITS.md for full attribution and docs/ARCHITECTURE.md/docs/MANDELBROT_ALGORITHM.md for how it works and what's planned.

Status: buildchain scaffolded, runs a placeholder test pattern in place of the real fractal generator -- see docs/MANDELBROT_ALGORITHM.md for the implementation plan.

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Building from source

Prerequisites

Tool	Purpose	Install
Oscar64	C cross-compiler targeting 6502/C64	build from source, see their README
wput	FTP deploy to the Ultimate device	<code>sudo apt install wput</code>
pandoc + texlive-xetex	Generate README.pdf (optional)	<code>sudo apt install pandoc texlive-xetex</code>

Deploy setup

Copy .env.example to .env and set ULTIP1 to your Ultimate device's IP address:

```
cp .env.example .env
```

Make targets

Target	Effect
make /make all	Compile to build/mandelupic.prg, regenerate README.pdf, build the release ZIP
make deploy	FTP the compiled .prg to the Ultimate device set in .env
make docs	Regenerate README.pdf via pandoc
make clean	Remove build outputs

Requires **Ultimate firmware 3.15 or newer** (UCI auto-enable, palette control commands -- see UCILIBMANUAL.md).